

Learning Behavior and AI Agency - Workshop on Predicting Performance from Reading and Learning Behavior and Orchestrating Human and AI Agency Joint Edition

Abstract: The expansion of digital learning materials has made rich behavioral data available, while the educational AI is shifting from reactive AI tools toward proactive, agentic AI systems. This convergence raises a central design question: how can agentic AI use behavioral data to deliver proactive interventions while preserving learner agency and teacher professional judgement? This joint edition at ICCE 2026 brings together the Workshop on Predicting Performance from Reading and Learning Behavior (9 editions at LAK) and HAI-Agency (AIED 2026) to address this question through three thematic areas: Learning Analytics for Performance Prediction, Agentic AI for Proactive and Reflective Learning, and, at their intersection, Behavior-Informed Agentic AI Design. Aligned with the ICCE 2026 theme of reimagining learning ecologies in the age of intelligent technologies, the workshop convenes researchers across learning analytics, AI in education, and human-AI interaction.

Keywords: Student Performance Prediction, Data Challenge, Learning Behavior, AI in Education, Agentic AI, Human-AI Interaction

1. Background

Over the past decade, Learning Analytics (LA) has established robust methodologies for analyzing digital learning traces. The use of digital textbooks and e-book systems has enabled large-scale analysis of reading behavior, including navigation, duration, and annotation patterns (Flanagan et al., 2018; Ogata et al., 2017). When combined with quiz scores, syllabus timelines, and lecture schedules, these behavioral data have enabled modeling of learning processes across preview, in-class, and review phases (Shimada et al., 2018), supporting at-risk identification, performance prediction, and evidence-based instructional interventions. Concurrently, Large Language Models and Generative AI have transformed AI in Education (AIED). Educational AI systems are evolving from reactive, prompt-based query tools into proactive Agentic AI systems (Lee & Palmer, 2025), shifting the role of AI from passive tool to active collaborator (Ogata et al., 2024, Chen et al., 2025; Li et al., 2026). However, the OECD (2026) cautions that AI systems can improve task performance without producing meaningful learning unless guided by pedagogical principles, defining a new design challenge at the intersection of AI autonomy and pedagogical alignment.

These two trajectories motivate a deeper integration of LA into research on human-AI interaction. Traditional LA has focused on structured learning logs such as reading behaviors, while next-generation analytics must examine dynamic human-AI interaction data including conversational processes, prompt refinement, help-seeking behaviors, and collaboration patterns. Such analyses can provide deeper insight into whether AI-supported environments truly facilitate meaningful learning processes.

2. Draft Call for Papers

2.1 Description

This one-day workshop focuses on the intersection of behavioral learning analytics and agentic AI in education, bridging ICCE's sub-conferences on **AIED/ITS (C1)** and **Advanced Learning Technologies and Learning Analytics (C3)**. There is growing interest in the learning analytics and educational data mining communities in extracting pedagogical signals from rich behavioral data, including reading patterns, navigation traces, and engagement logs from digital textbooks and learning platforms, to predict performance, detect engagement changes, and provide actionable feedback to learners and teachers. Simultaneously, the AIED community is shifting from reactive AI tools toward proactive, teammate-like AI systems, raising a central design challenge: how can agentic AI use behavioral data to deliver proactive interventions while preserving learner agency and teacher professional judgement?

2.2 Topics of Interest

We welcome submissions on (but not limited to) the following topics:

Learning Analytics for Performance Prediction

- Student performance/at-risk prediction
- Student reading behavior self-regulation profiles spanning the entire course
- Preview, in-class, and review reading patterns
- Student engagement analysis; and behavior change detection
- Visualization methods to inform and provide meaningful feedback to stakeholders

Agentic AI for Proactive and Reflective Learning

- Proactive and reflective learning, co-regulation, and learner agency
- Proactive/reflective prompts, scaffolds, and personalized interactions
- Teacher-facing design and human-AI co-orchestration
- Explainable, teachable, and designable AI systems
- Process-based and human-AI collaborative assessment

Behavior-Informed Agentic AI Design

- Behavioral signals triggering proactive AI interventions
- Adaptive scaffolding informed by longitudinal learning traces
- Predictive models for calibrating agentic interventions
- Process mining of human-AI interaction
- Detecting AI overreliance and productive co-learning through interaction signals

We additionally invite submissions using the learning behavior dataset and the OpenLA library. Dataset access requires a brief application; details will be published on the workshop website.

2.3 Submission instructions

The workshop invites the submission of full, short, and position papers following the ICCE 2026 paper template. All submissions undergo single-blind review by at least two program committee members, assessed for quality, rigor, originality, relevance to the workshop scope, and ethical consideration.

3. Organization

3.1 Format and Logistics

Format and Logistics The workshop is a full-day in-person mini-conference at ICCE 2026, comprising a joint keynote, paper presentations grouped by the three thematic areas, and a concluding discussion on open problems and future directions. The discussion invites participants to identify emerging research challenges and collaboration opportunities across the two communities. Expected attendance is 10–20 participants, drawn from learning analytics, AI in education, and human-AI interaction.

3.2 Workshop organizers

Dr. Gen LI (gen.li@limu.ait.kyushu-u.ac.jp) is an Assistant Professor at Kyushu University, working on AI in Education, intelligent tutoring systems, and agent systems. He serves on the LAK program committee and as Publication Co-Chair of PRICAI 2026.

Dr. Li CHEN (chen-l68@cc.osaka-kyoiku.ac.jp) is a lecturer at Osaka Kyoiku University. She serves as an active program committee member for leading conferences (LAK, AIED), being awarded the best paper award at CELDA2020. Her research interests include learning analytics and STEM education.

Dr. Boxuan MA (boxuan@artsci.kyushu-u.ac.jp) is an Assistant Professor at Kyushu University. His research spans AI in Education, educational data mining, learning analytics, and human-computer interaction; he serves on the program committees of LAK, EDM, and AIED.

Dr. Huiyong LI (li.huiyong.194@m.kyushu-u.ac.jp) is an Assistant Professor at Kyushu University, and Virtual Experience Co-Chair for AIED 2026. His research focuses on learning analytics and AI in education.

Dr. Yiling DAI (daiyl@hiroshima-u.ac.jp) is an Assistant Professor at Hiroshima University, working on educational data mining, learning analytics, and explainable AI-driven recommendations. She serves on the program committees of LAK, ICCE, and AIED, and received the best reviewer award at ICCE 2024.

Prof. Brendan FLANAGAN (flanagan@fc.ritsumei.ac.jp) is a Professor at Ritsumeikan University working on AI in Education, learning analytics, NLP/Text Mining, and Computer-Assisted Language Learning. He hosted

and chaired LAK 2024 in Kyoto and has organized workshops at LAK, AIED, and ICCE; he is principal investigator on multiple national research projects on educational symbiotic AI systems.

Prof. Daisuke DEGUCHI (ddeguchi@nagoya-u.ac.jp) is an Associate Professor at Nagoya University. His research focuses primarily on computer vision and pattern recognition, and he is working on the development of AI for educational applications using these technologies.

Prof. Takayoshi YAMASHITA (takayoshi@isc.chubu.ac.jp) is a Professor at Chubu University. His research focuses on computer vision, machine learning, and learning analytics, particularly in human understanding, autonomous driving, and bioinformatics. He is also conducting research on grade prediction and its explainability in educational applications.

Prof. Atsushi SHIMADA (atsushi@ait.kyushu-u.ac.jp) is a Professor at Kyushu University, working on learning analytics, AI in education, pattern recognition, and media processing. He has published over 80 journal papers and 300 international conference papers, and received awards such as the IPSJ/IEEE Computer Society Young Computer Researcher Award and the MEXT Young Scientist Award.

Prof. Hiroaki OGATA (ogata.hiroaki.3e@kyoto-u.ac.jp) is the current President of APSCE and a Professor and Director at the Academic Center for Computing and Media Studies, Kyoto University. He has published over 700 peer-reviewed papers and received the APSCE Distinguished Researcher Award.

3.3 Prospective Members of Program Committee

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